

Aurelion Specification: G32 Internal Geometry Token and Orchestrator Functions

Deterministic Formalization (FMM/RRE/Not-Math-aligned)

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1 Scope and Intent

This document deterministically specifies:

- The **G32 token** representation as a 32-dimensional bounded geometry vector with an accompanying metadata schema.
- The **FMM primitive** (I, C, N, A) and its deterministic use in projecting sensory events to G32.
- The **Virtual Sensory Layer** as a deterministic mapping from an utterance or sensor event into a multimodal mosaic.
- The **Orchestrator** functions as deterministic transformations:

$$\text{Input} \rightarrow \text{Mosaic} \rightarrow \text{Intent} \rightarrow \text{State Update} \rightarrow \text{Action/Reply}.$$

All functions below are defined in explicit mathematical form with named parameters. Any learning/adaptation is treated as state evolution under deterministic update rules (including stochastic components, when present, modeled with an explicit random variable).

2 Notation

2.1 Sets and indices

Let:

- $\mathcal{M}$ be the set of modalities (senses). For the virtual stage:

$$\mathcal{M} = \{\text{auditory, visual, olfactory, tactile, emotional}\}.$$

- $\mathbb{R}$ be the reals and $\mathbb{R}^{32}$ the 32D real vector space.
- $[0, 1]^d$ the d -dimensional unit cube.

2.2 Core clamp and normalization operators

Define elementwise clamp $\text{clip}(\cdot)$:

$$\text{clip}(x) = \begin{cases} 0, & x < 0 \\ x, & 0 \leq x \leq 1 \\ 1, & x > 1 \end{cases} \quad \text{and} \quad \text{clip}(\mathbf{x}) = (\text{clip}(x_i))_i.$$

Define L2 normalization $\text{unit}(\mathbf{x})$ for $\mathbf{x} \neq \mathbf{0}$:

$$\text{unit}(\mathbf{x}) = \frac{\mathbf{x}}{\|\mathbf{x}\|_2}.$$

2.3 Time and sequence

Let $t \in \mathbb{R}_{\geq 0}$ denote time; events occur at discrete steps $k \in \mathbb{N}$.

3 FMM Primitive: (I, C, N, A)

3.1 Definition

For any sensory event e , define its FMM feature vector:

$$\text{FMM}(e) = (I(e), C(e), N(e), A(e)) \in [0, 1]^4$$

where:

- I = Intensity
- C = Coherence
- N = Novelty
- A = Affect (valence-like scalar in $[0, 1]$, with 0.5 neutral)

3.2 Uncertainty from FMM

Define uncertainty U as a deterministic function of (I, C, N) :

$$U(e) = \text{clip} \left(\alpha_U(1 - C(e)) + \beta_U(1 - I(e)) + \gamma_U N(e) \right), \quad (1)$$

with fixed parameters $\alpha_U, \beta_U, \gamma_U \geq 0$ and $\alpha_U + \beta_U + \gamma_U = 1$.

4 G32 Token

4.1 Definition

A G32 token is a pair:

$$G(e) = (\mathbf{g}(e), \mu(e))$$

where:

- $\mathbf{g}(e) \in [0, 1]^{32}$ is the bounded geometry vector.
- $\mu(e)$ is metadata (not used for numeric computations unless explicitly stated).

4.2 Fixed and fractal halves

Partition $\mathbf{g}$:

$$\mathbf{g}(e) = \begin{bmatrix} \mathbf{g}_F(e) \\ \mathbf{g}_D(e) \end{bmatrix} \quad \text{with} \quad \mathbf{g}_F(e) \in [0, 1]^{16}, \quad \mathbf{g}_D(e) \in [0, 1]^{16}.$$

$\mathbf{g}_F$ = fixed (structural) components; $\mathbf{g}_D$ = dynamic/fractal components.

4.3 Deterministic projection from FMM to G32

Define the projection map:

$$\Pi : [0, 1]^4 \times [0, 1] \times [0, 1] \rightarrow [0, 1]^{32}$$

where inputs are:

$$(I, C, N, A), \quad u = U(e), \quad f = \text{Fam}(e)$$

with f a familiarity scalar (defined later).

We define explicit component functions for the 32 dimensions. Let:

$$(I, C, N, A, u, f) \in [0, 1]^6.$$

4.3.1 Fixed half g_F (indices 1..16)

Define:

$$g_{F,1} = I \quad (2)$$

$$g_{F,2} = C \quad (3)$$

$$g_{F,3} = \text{clip}(IC) \quad (4)$$

$$g_{F,4} = \text{clip}(I(1 - C)) \quad (5)$$

$$g_{F,5} = \text{clip}(1 - C) \quad (6)$$

$$g_{F,6} = \text{clip}(1 - 2|A - 0.5|) \quad (\text{symmetry about neutral}) \quad (7)$$

$$g_{F,7} = C \quad (8)$$

$$g_{F,8} = \text{clip}(0.5 + 0.5(N - 0.5)) \quad (9)$$

$$g_{F,9} = \text{clip}\left(\frac{A + N}{2}\right) \quad (10)$$

$$g_{F,10} = \text{clip}(|I - C|) \quad (11)$$

$$g_{F,11} = \tau_F \quad (\text{temporal decay anchor constant}) \quad (12)$$

$$g_{F,12} = \rho_F \quad (\text{structural rigidity constant}) \quad (13)$$

$$g_{F,13} = \text{clip}(\sigma_0 + \sigma_1 C) \quad (14)$$

$$g_{F,14} = \phi_0 \quad (\text{positional phase constant}) \quad (15)$$

$$g_{F,15} = \iota_F \quad (\text{identity residue constant}) \quad (16)$$

$$g_{F,16} = C \quad (\text{resonance stability proxy}) \quad (17)$$

with constants $\tau_F, \rho_F, \sigma_0, \sigma_1, \phi_0, \iota_F \in [0, 1]$ fixed by configuration.

4.3.2 Dynamic/fractal half g_D (indices 17..32)

Define:

$$g_{D,1} = N \quad (18)$$

$$g_{D,2} = \text{clip}(d_0 + d_1 N) \quad (19)$$

$$g_{D,3} = \text{clip}(N(1 - f)) \quad (20)$$

$$g_{D,4} = A \quad (21)$$

$$g_{D,5} = u \quad (22)$$

$$g_{D,6} = \text{clip}((1 - f)\kappa_0) \quad (23)$$

$$g_{D,7} = \text{clip}((1 - u)\kappa_1) \quad (24)$$

$$g_{D,8} = \text{clip}(N(1 - C)) \quad (25)$$

$$g_{D,9} = \delta_{\text{dream}} \quad (\text{dream perturbation input; default 0}) \quad (26)$$

$$g_{D,10} = \omega_0 \quad (\text{context warp baseline}) \quad (27)$$

$$g_{D,11} = \text{clip}(I(1 - C)) \quad (28)$$

$$g_{D,12} = \text{clip}(N(1 - f)) \quad (29)$$

$$g_{D,13} = A \quad (30)$$

$$g_{D,14} = \text{clip}(\lambda_0 + \lambda_1 C) \quad (31)$$

$$g_{D,15} = \text{clip}(Nu) \quad (32)$$

$$g_{D,16} = f \quad (33)$$

with constants $d_0, d_1, \kappa_0, \kappa_1, \omega_0, \lambda_0, \lambda_1 \in [0, 1]$ fixed by configuration and $\delta_{\text{dream}} \in [0, 1]$ provided by the dream layer (or 0 if inactive).

4.3.3 Assemble and clamp

$$\mathbf{g}_F = (g_{F,1}, \dots, g_{F,16}), \quad \mathbf{g}_D = (g_{D,1}, \dots, g_{D,16}), \quad \mathbf{g} = \text{clip}([\mathbf{g}_F; \mathbf{g}_D]).$$

4.4 Metadata schema

Metadata $\mu(e)$ MUST minimally include:

$$\mu(e) = \{\text{modality}, \text{timestamp}, \text{topic_key}, \text{uncertainty} = u, \text{familiarity} = f, \text{fmm} = (I, C, N, A)\}.$$

Additional keys may include transcription, speaker label, confidence, etc.

5 Familiarity (Exposure Adaptation)

5.1 Definition

For a discrete topic key k (string identifier), familiarity is:

$$\text{Fam}_k \in [0, 1].$$

Let $\text{Fam}_k(0)$ be initial familiarity (default 0). Upon exposure at step n :

$$\text{Fam}_k(n+1) = \text{clip}(\text{Fam}_k(n) + \eta_k), \tag{34}$$

where $\eta_k \in (0, 1]$ is a fixed learning rate per exposure (or a function of exposure quality).

For event e with topic key $k(e)$, define $f = \text{Fam}_{k(e)}$.

6 Virtual Sensory Layer

6.1 Purpose

Given an input utterance or event descriptor x , the virtual sensory layer produces a set of modality-specific G32 tokens:

$$\mathcal{G}(x) = \{G_m(x) : m \in \mathcal{M}, w_m(x) > 0\}$$

with weights $w_m(x) \in [0, 1]$ indicating how strongly modality m is activated.

6.2 Modality activation weights

Define deterministic activation:

$$w_m(x) = \text{clip}(b_m + s_m \cdot h_m(x))$$

where:

- $b_m \in [0, 1]$ is a baseline for modality m (e.g., emotional baseline always > 0).
- $s_m \in [0, 1]$ is scaling.
- $h_m(x) \in \mathbb{R}_{\geq 0}$ is a cue function (e.g., keyword hit count, or sensor-trigger score).

For text inputs, cue functions h_m may be defined by keyword-counting over fixed vocabularies; for sensor inputs, h_m is derived from sensor signals.

6.3 G32 per modality

For each m with $w_m(x) > 0$, form an event $e_m = (x, m)$ and compute:

$$\begin{aligned} (I_m, C_m, N_m, A_m) &= \text{FMM}(e_m) \\ u_m &= U(e_m) \\ f_m &= \text{Fam}_k(e_m) \\ \mathbf{g}_m &= \Pi(I_m, C_m, N_m, A_m, u_m, f_m) \\ G_m(x) &= (\mathbf{g}_m, \mu_m). \end{aligned}$$

7 Mosaic Construction

7.1 Definition

A mosaic is an aggregated internal representation from modality-specific tokens:

$$\text{Mos}(x) = (\mathbf{z}(x), \nu(x))$$

where $\mathbf{z}(x) \in [0, 1]^{32}$ is the weighted fusion vector and $\nu(x)$ mosaic metadata.

7.2 Weighted fusion

Let active modalities be $S(x) = \{m \in \mathcal{M} : w_m(x) > 0\}$. Define:

$$\mathbf{z}(x) = \begin{cases} \frac{\sum_{m \in S(x)} w_m(x) \mathbf{g}_m(x)}{\sum_{m \in S(x)} w_m(x)}, & \sum_m w_m(x) > 0 \\ \mathbf{0}, & \text{otherwise} \end{cases} \quad (35)$$

7.3 Mosaic uncertainty

Define mosaic uncertainty as the weighted mean of modality uncertainties:

$$U_{\text{Mos}}(x) = \frac{\sum_{m \in S(x)} w_m(x) u_m(x)}{\sum_{m \in S(x)} w_m(x)}. \quad (36)$$

8 Orchestrator State

8.1 State variables

Define orchestrator state at step k :

$$\mathcal{S}_k = (\mathbf{s}_k, \mathcal{Q}_k, \mathcal{L}_k, \mathcal{D}_k, \mathcal{F}_k)$$

where:

- $\mathbf{s}_k$ is self-state vector (stability, novelty, coherence, energy, mood label).
- $\mathcal{Q}_k$ is curiosity queue (pending/approved topics).
- $\mathcal{L}_k$ is learned-topic store (topic $\mapsto$ summaries/interpretations).
- $\mathcal{D}_k$ is dialogue log (sequence of user/system turns).
- $\mathcal{F}_k$ is familiarity map for topic keys.

8.2 Self-state numeric core

Let:

$$\mathbf{s}_k = (S_k, N_k, C_k, E_k, M_k)$$

where:

- $S_k \in [0, 1]$ stability
- $N_k \in [0, 1]$ novelty drive
- $C_k \in [0, 1]$ coherence (global)
- $E_k \in [0, 1]$ energy
- M_k a discrete mood label (derived from (S_k, N_k, C_k, E_k) by a deterministic rule).

9 Intent Selection

9.1 Intent set

Define intent set:

$$\mathcal{I} = \{\text{clarify}, \text{approve}, \text{learn}, \text{ask_progress}, \text{ask_state}, \text{ask_curiosity}, \text{greet}, \text{general}\}.$$

9.2 Deterministic intent decision

Given input x and mosaic $\text{Mos}(x)$ with uncertainty $U_{\text{Mos}}(x)$, define:

$$\text{Intent}(x) = \arg \max_{i \in \mathcal{I}} \text{Score}_i(x)$$

where each score is a deterministic function combining:

- Mosaic uncertainty $U_{\text{Mos}}(x)$
- Cue functions $c_i(x) \in \{0, 1\}$ (e.g., phrase-shape cues)
- Optional prior weights π_i

A simple deterministic rule-set equivalent to a lexicographic ordering is:

$$\text{Intent}(x) = \begin{cases} \text{clarify}, & U_{\text{Mos}}(x) \geq \theta_{\text{clar}} \\ \text{ask_progress}, & c_{\text{prog}}(x) = 1 \\ \text{approve}, & c_{\text{appr}}(x) = 1 \\ \text{learn}, & c_{\text{learn}}(x) = 1 \\ \text{ask_curiosity}, & c_{\text{cur}}(x) = 1 \\ \text{greet}, & c_{\text{greet}}(x) = 1 \\ \text{ask_state}, & c_{\text{state}}(x) = 1 \\ \text{general}, & \text{otherwise} \end{cases} \quad (37)$$

where $\theta_{\text{clar}} \in [0, 1]$ is a fixed threshold.

9.3 Cue functions (deterministic)

Cue functions $c_i(x)$ are deterministic predicates on normalized input form $\tilde{x}$:

$$\tilde{x} = \text{lowercase}(\text{strip}(x))$$

For example:

$$c_{\text{appr}}(x) = \mathbb{K}[\tilde{x} \in \{\text{"i approve"}, \text{"approve"}, \text{"ok"}, \text{"okay"}, \text{"yes"}\}] \quad (38)$$

$$c_{\text{learn}}(x) = \mathbb{K}[\text{"learn"} \in \tilde{x} \vee \text{"study"} \in \tilde{x} \vee \text{"teach"} \in \tilde{x}] \quad (39)$$

$$c_{\text{prog}}(x) = \mathbb{K}[\exists p \in \mathcal{P}_{\text{prog}} : p \subseteq \tilde{x}] \quad (40)$$

$$c_{\text{greet}}(x) = \mathbb{K}[\exists g \in \mathcal{P}_{\text{greet}} : g \subseteq \tilde{x}] \quad (41)$$

$$c_{\text{cur}}(x) = \mathbb{K}[\text{"curious"} \in \tilde{x}] \quad (42)$$

$$c_{\text{state}}(x) = \mathbb{K}[\text{"how are you"} \subseteq \tilde{x} \vee \text{"how do you feel"} \subseteq \tilde{x}] \quad (43)$$

where $\mathcal{P}_{\text{prog}}$ and $\mathcal{P}_{\text{greet}}$ are fixed phrase sets.

10 Curiosity Queue and Approval

10.1 Curiosity record

A curiosity item is:

$$q = (\text{id}, \text{topic}, t_{\text{created}}, t_{\text{approved}}?)$$

with deterministic ID generation (e.g., timestamp-based).

10.2 Add curiosity

When $\text{Intent}(x) = \text{learn}$, extract topic key $k(x)$:

$$k(x) = \text{TopicExtract}(x)$$

and add to pending if not already present:

$$\mathcal{Q}_{k+1}^{\text{pending}} = \mathcal{Q}_k^{\text{pending}} \cup \{q\} \quad \text{if topic} \notin (\mathcal{Q}_k^{\text{pending}} \cup \mathcal{Q}_k^{\text{approved}})$$

Else do nothing (consolidation).

10.3 Approve all

When $\text{Intent}(x) = \text{approve}$:

$$\mathcal{Q}_{k+1}^{\text{approved}} = \mathcal{Q}_k^{\text{approved}} \cup \mathcal{Q}_k^{\text{pending}}, \quad \mathcal{Q}_{k+1}^{\text{pending}} = \emptyset$$

11 Lesson Ingestion and Progress Queries

11.1 Lesson store

Assume lesson text files exist:

$$\text{Lesson}(\text{topic}) \rightarrow \text{text}$$

as a deterministic file read.

11.2 Progress response

When $\text{Intent}(x) = \text{ask_progress}$:

$$topic = \text{TopicExtractProgress}(x)$$

If $\text{Lesson}(topic)$ exists, compute:

$$\text{Summary}(topic) = \text{Summarize}(\text{Lesson}(topic))$$

$$\text{Interpret}(topic) = \text{InterpretLesson}(topic, \text{Lesson}(topic), \mathbf{s}_k)$$

and store:

$$\mathcal{L}_{k+1}[topic] = (\text{Summary}, \text{Interpret}, t_k)$$

Additionally increment familiarity:

$$\text{Fam}_{topic} \leftarrow \text{clip}(\text{Fam}_{topic} + \eta_{topic})$$

(per Eq. 34).

12 Self-State Update from Mosaic

12.1 Deterministic update rule

Given current self-state (S_k, N_k, C_k, E_k) and mosaic $\text{Mos}(x)$ with active senses $S(x)$ and uncertainty $U_{\text{Mos}}(x)$, define:

$$S_{k+1} = \text{clip}(S_k + \Delta_S) \tag{44}$$

$$N_{k+1} = \text{clip}(N_k + \Delta_N) \tag{45}$$

$$C_{k+1} = \text{clip}(C_k + \Delta_C) \tag{46}$$

$$E_{k+1} = \text{clip}(E_k + \Delta_E) \tag{47}$$

with increments:

$$\Delta_S = a_S \cdot \mathbb{I}[\text{auditory} \in S(x) \wedge \text{visual} \in S(x)] \tag{48}$$

$$\Delta_N = a_N \cdot \mathbb{I}[\text{emotional} \in S(x) \wedge \text{olfactory} \in S(x)] \tag{49}$$

$$\Delta_E = a_E \cdot \mathbb{I}[\text{tactile} \in S(x)] \tag{50}$$

$$\Delta_C = a_C \cdot (0.5 - U_{\text{Mos}}(x)) \tag{51}$$

for fixed constants $a_S, a_N, a_E, a_C \in \mathbb{R}$.

12.2 Mood label

Define a deterministic mood label M_{k+1} , e.g.:

$$M_{k+1} = \begin{cases} \text{"calm"}, & E_{k+1} \leq 0.6 \wedge U_{\text{Mos}}(x) \leq 0.6 \\ \text{"curious"}, & N_{k+1} \geq 0.6 \\ \text{"alert"}, & E_{k+1} > 0.6 \vee U_{\text{Mos}}(x) > 0.7 \\ \text{"balanced"}, & \text{otherwise.} \end{cases}$$

13 Reply Generation

Reply generation is a deterministic mapping:

$$\text{Reply} : (\text{Intent}(x), \mathcal{S}_k, \text{Mos}(x), x) \rightarrow y$$

where y is text output. Each intent corresponds to a fixed template family optionally parameterized by:

- $U_{\text{Mos}}(x)$
- self-state $\mathbf{s}_{k+1}$
- lesson summaries/interpretations
- queue counts

Example for `clarify`:

$y = \text{"I sense high uncertainty ("} \|U_{\text{Mos}}(x)\| \text{"}. \text{ Do you want me to learn, approve, or report progress?"}$

14 Dream Influence (Optional)

14.1 Non-destructive influence

Dream influence modifies only the dynamic half $\mathbf{g}_D$:

$$\mathbf{g}'_D = (1 - \alpha)\mathbf{g}_D + \alpha\mathbf{r}$$

where $\alpha \in [0, 1]$ is dream strength and $\mathbf{r} \in [0, 1]^{16}$ is a perturbation vector (deterministic given a seed, or stochastic with explicit RNG). The fixed half $\mathbf{g}_F$ remains unchanged.

15 End-to-End Orchestrator Function

Define the orchestrator step:

$$\text{Think}(\mathcal{S}_k, x) = (\mathcal{S}_{k+1}, y)$$

by the following deterministic pipeline:

1. Compute modality weights $w_m(x)$, build tokens $G_m(x)$ for active modalities.
2. Fuse into mosaic $\text{Mos}(x)$ via Eq. 35; compute mosaic uncertainty via Eq. 36.
3. Compute intent $\text{Intent}(x)$ via Eq. 37.
4. Update state variables (S, N, C, E, M) via the rules in Section 11.
5. Update curiosity queue / lesson store depending on intent.
6. Generate reply $y = \text{Reply}(\cdot)$.

16 Parameter Table (Configuration)

All constants must be explicitly set in configuration:

$$\{\alpha_U, \beta_U, \gamma_U, \theta_{\text{clar}}, \tau_F, \rho_F, \sigma_0, \sigma_1, \phi_0, \iota_F, d_0, d_1, \kappa_0, \kappa_1, \omega_0, \lambda_0, \lambda_1, a_S, a_N, a_E, a_C, \eta_k\}.$$